

# SAWEC: Sensing-Assisted Wireless Edge Computing



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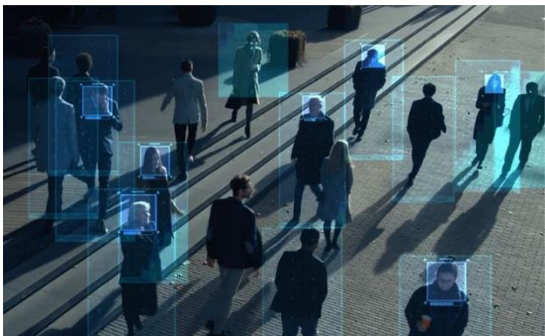
# AR/VR: Next Frontier in Pervasive Computing

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We are **very far** from *truly seamless*  
and *immersive* AR/VR experiences

# What is the current bottleneck?



Semantic  
Segmentation

Object  
Recognition

Compute-Intensive  
AI/ML Tasks

120 Hz frame rate  
and 8K resolution  
are needed!

# State of current wireless AR/VR headsets

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Meta Quest 3  
1.5-2 hours

**Weight**  
515 grams

**Refresh Rate**  
Up to 90 Hz

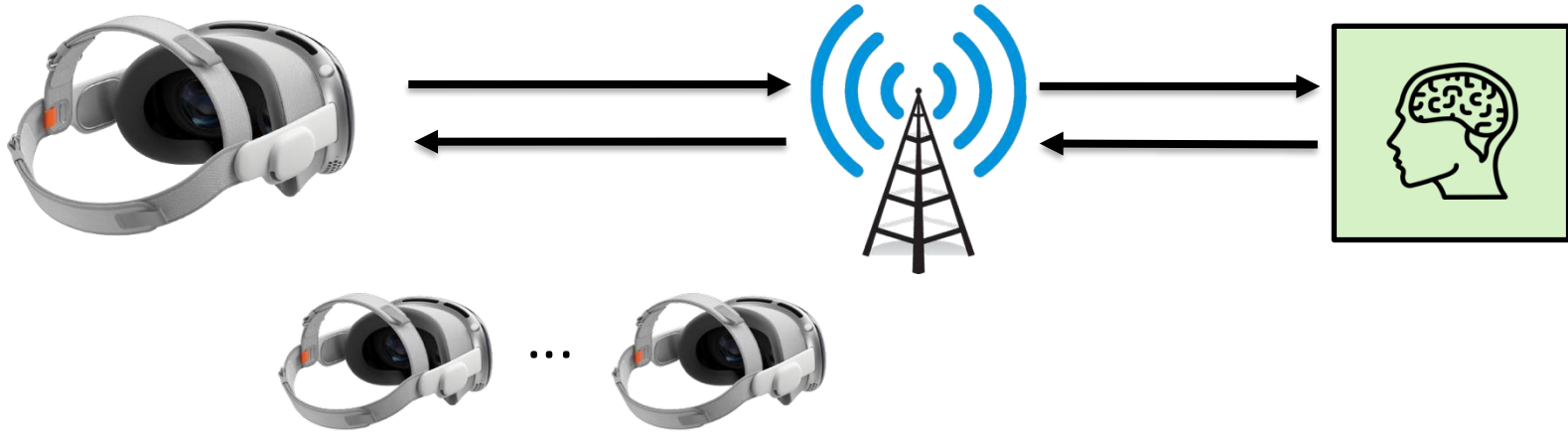


Apple Vision Pro  
Up to 2 hours

**Weight**  
650 grams plus  
353 grams battery

**Refresh Rate**  
Up to  
90 Hz

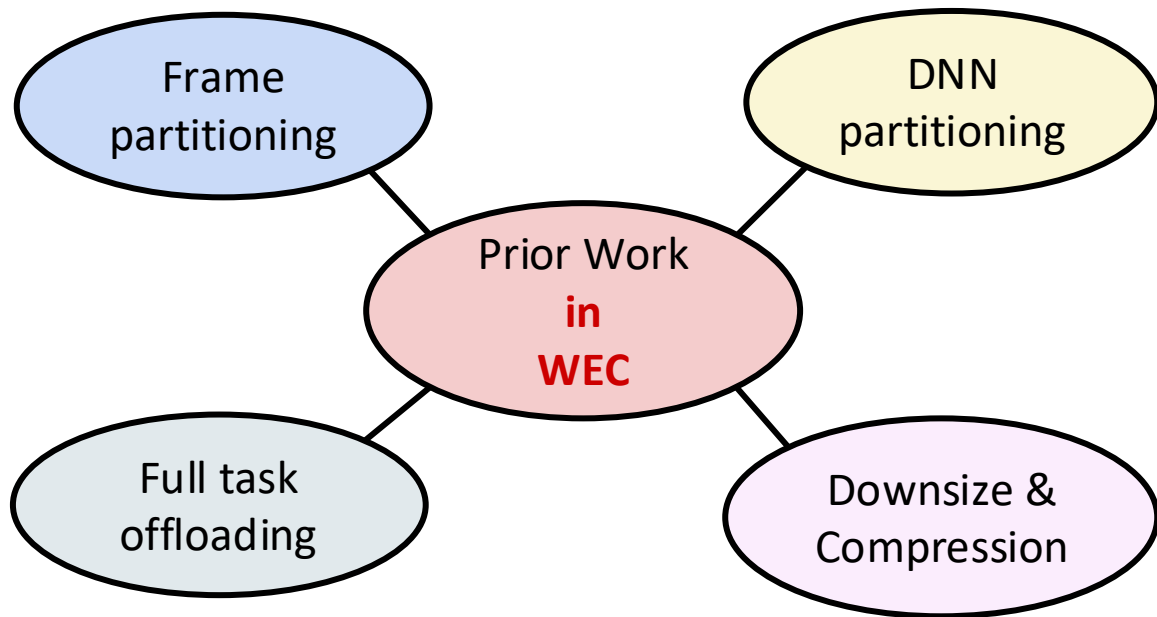
120 Hz  $\rightarrow$  E2E latency of 8 milliseconds



120 Hz @ 8K  $\rightarrow$  40 Gbps  $\rightarrow$  Wi-Fi supports 1.2 Gbps

# What Has Been Done So Far?

- Z. Yang et al., “Edgeduet: Tiling Small Object Detection for Edge Assisted Autonomous Mobile Vision,” IEEE/ACM Transactions on Networking, 2022.
- S. Laskaridis et al., “SPINN: Synergistic Progressive Inference of Neural Networks Over Device and Cloud,” in Proceedings of MobiCom, 2020, pp. 1–15.
- R. Xu et al., “ApproxDet: Content and Contention-Aware Approximate Object Detection for Mobiles,” in Proceedings of SenSys, 2020, pp. 449–462
- W. Jiang, Z. Teng, M. Li, and L. P. Qian, “Energy-Efficient Data Compression and Resource Allocation for Edge Assisted IoT Networks,” in Proc. of ICC, IEEE, 2023, pp. 1–6.



# Does Compression Degrade Performance?

Happy Angry Neutral Surprise

1024 x 1024



128 X 128



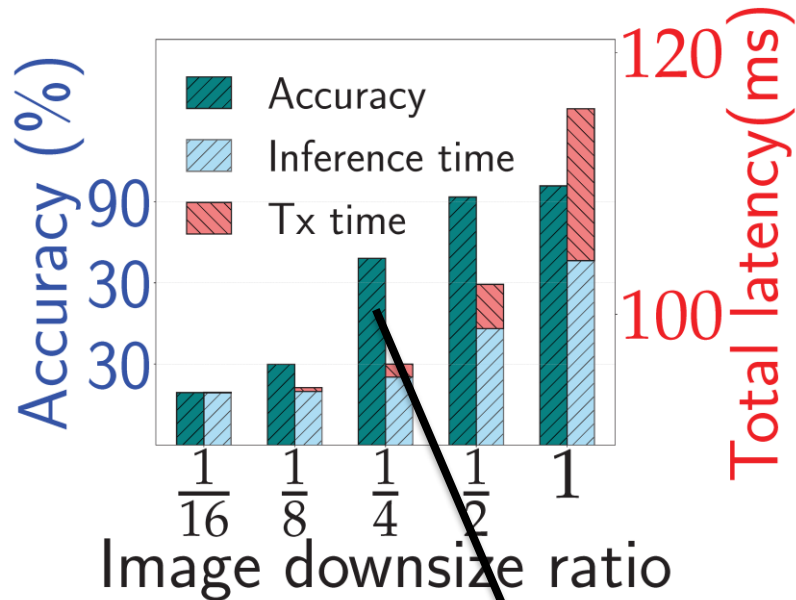
32 X 32



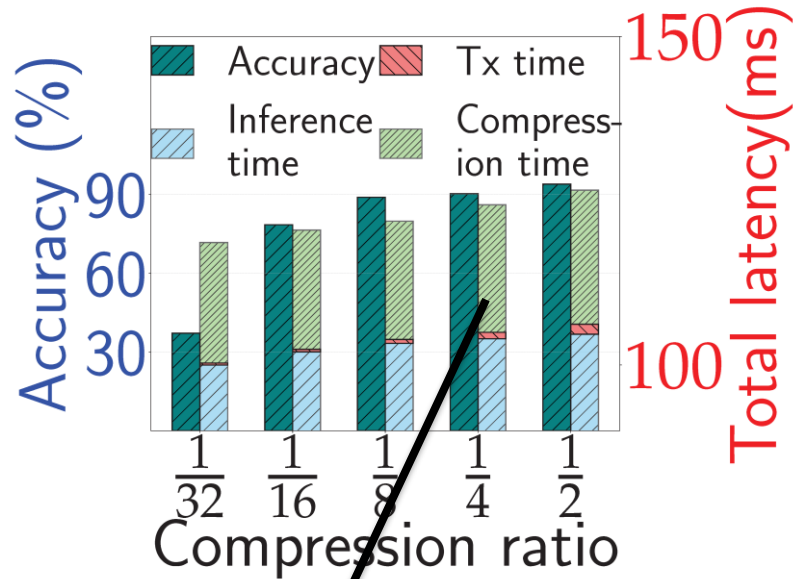
Six facial emotion classes from six different subjects



# Does Compression Degrade Performance?



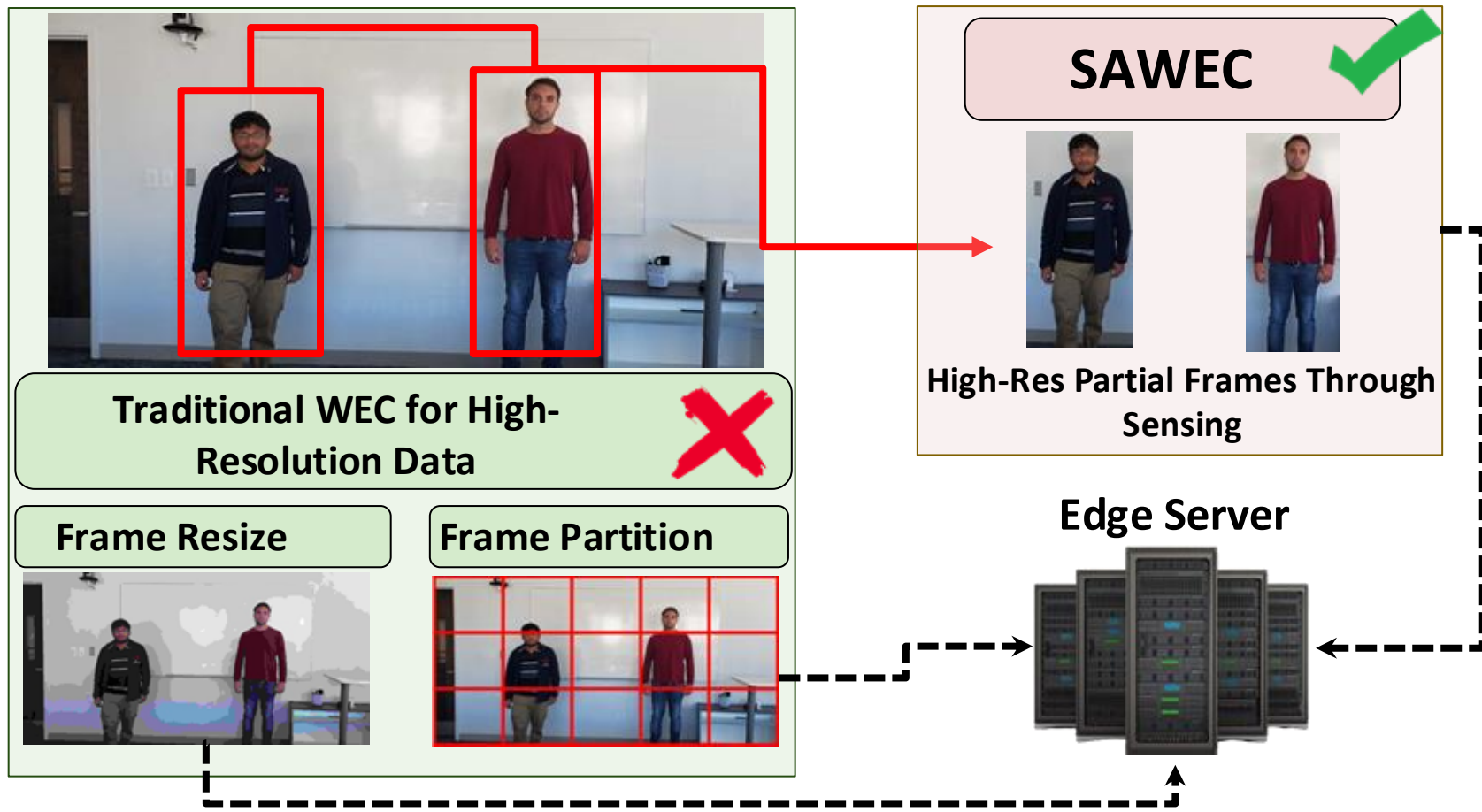
Up to 50% drop with 1/4



~30 ms compression time!



# SAWEC: Sensing-Assisted Wireless Edge Computing



# Summary of Key Contributions

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- ✓ We propose SAWEC, a new paradigm for wireless edge computing
- ✓ E2E latency is optimized by offloading only the relevant data
- ✓ SAWEC reduces the channel occupation and E2E latency by 94% and 96%
- ✓ Improves the instance segmentation performance by 47%

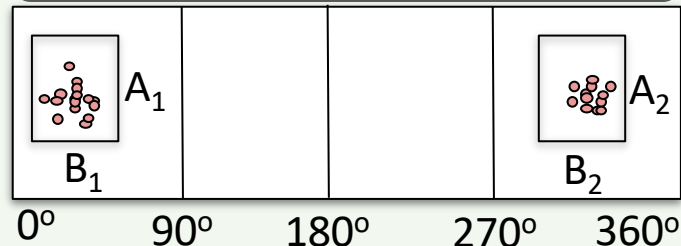
# Overview of SAWEC

1

## Sensing-Assisted ROI Detection



Find ROI with Wireless Sensing



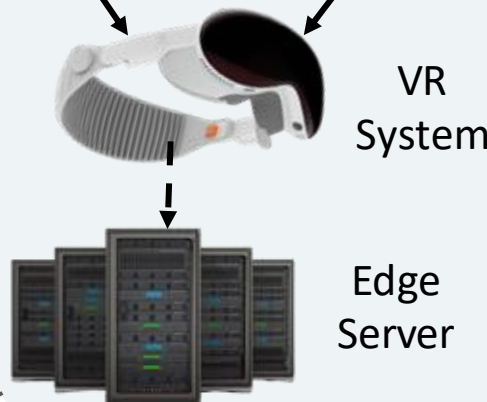
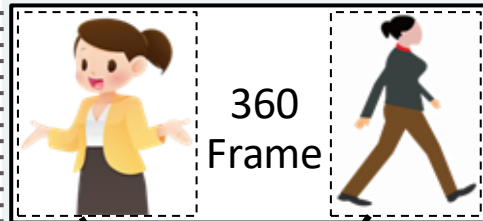
Find  $A_x, B_x$  For all  $ROI_N$

$N = 1, 2, 3, \dots, n$

2

## ROI Offloading

$ROI_1 \dots \dots \dots ROI_N$

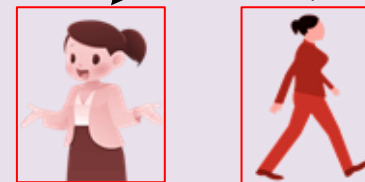


3

## Task Execution at Server



ML  
Task



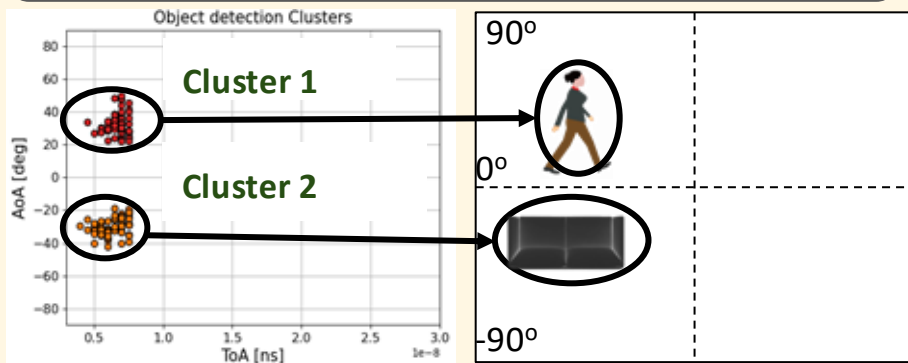
talking  
avatar

walking  
avatar

# Detection of the Region of Interest (ROI)

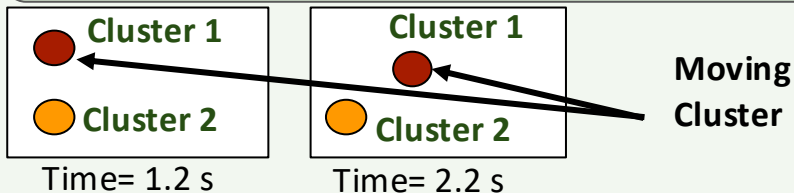
1

Localization based on AoA and ToA



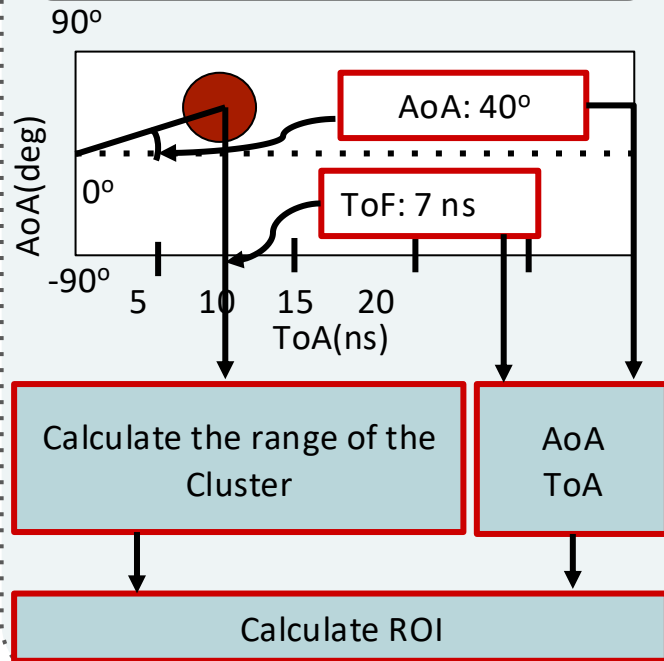
2

Track Environmental Change  
Through Moving Cluster

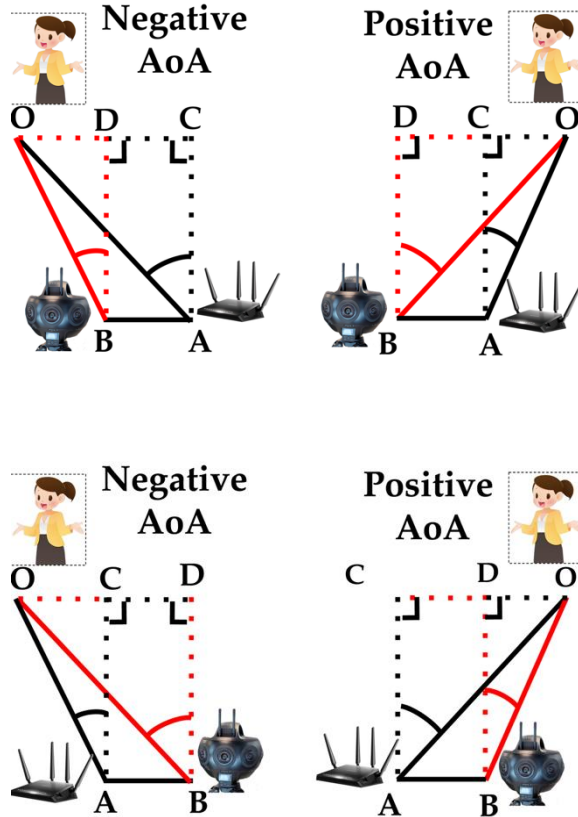


3

Find ROI based on AoA, ToA and  
Cluster Radius



# Cluster Projection to Camera Ref. System (1)




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**Algorithm 1** AoA and ToA projection into the camera reference

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**Require:**  $A_x, B_x, \angle OAC, OA$

**Ensure:**  $\angle OBD, OB$

$$OC = OA \cdot \sin(\angle OAC)$$

$$CD = AB$$

$$BD = AC = OA \cdot \cos(\angle OAC)$$

$$\zeta_{\text{pos}} \leftarrow \text{sgn}(A_x - B_x) \quad \triangleright \zeta_{\text{pos}} = 1 \text{ if camera is on the left of Wi-Fi NIC}$$

$$\zeta_{\text{aoa}} \leftarrow \text{sgn}(\angle OAC) \quad \triangleright \zeta_{\text{aoa}} = 1 \text{ if positive AoA}$$

$$OD = OC + \zeta_{\text{pos}} * \zeta_{\text{aoa}} * CD$$

$$\angle OBD = \arctan\left(\frac{OD}{BD}\right); \quad OB = \left(\frac{OD}{\sin \angle OBD}\right)$$

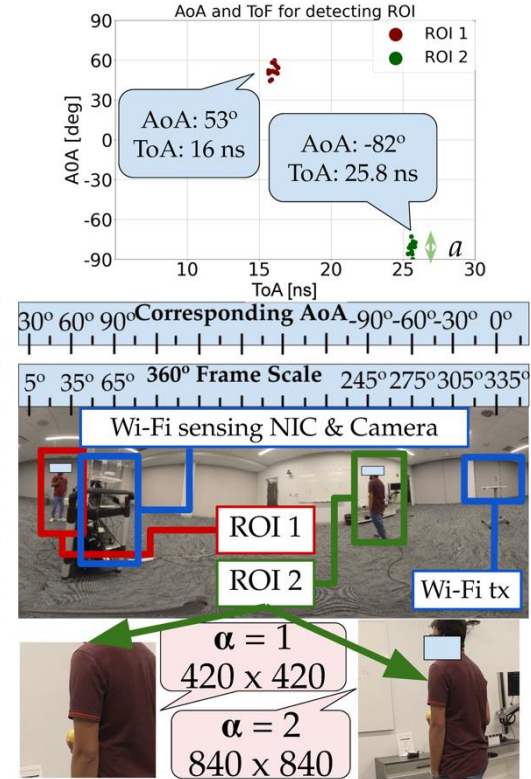
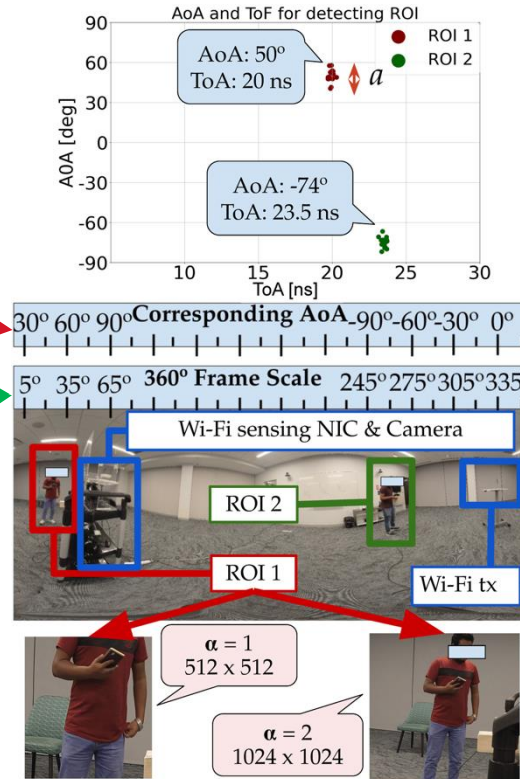

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# Cluster Projection to Camera Ref. System (2)

●  $\theta = [(\angle OBD + \theta_{tx}) \bmod 360]$

Where,

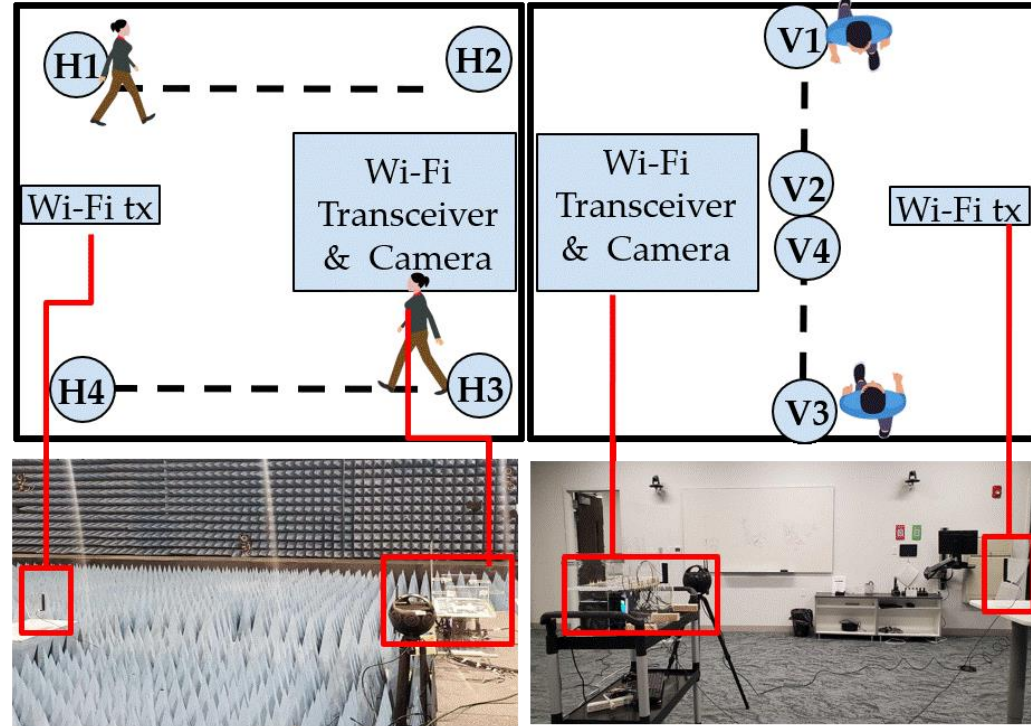
- $\angle OBD$  is **AoA** for camera ref. system
- $\theta$  is the object in 360 frame scale
- $\theta_{tx}$  is **Wi-Fi tx** in 360 frame scale
- mod represents modulo operation





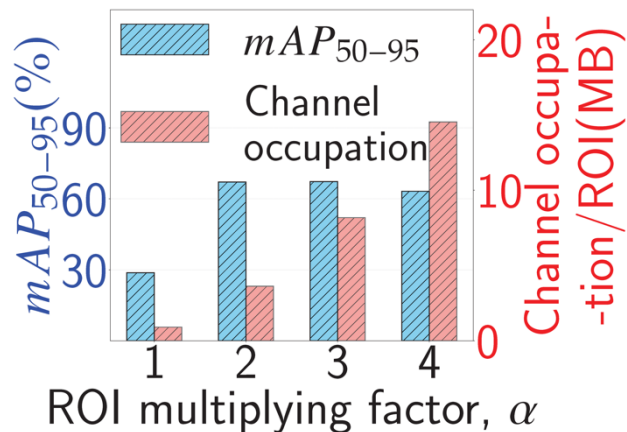
# Experimental Setup

- Components:
  - Insta 360 camera
  - Co-located IEEE 802.11ax client operating at 160 MHz
- PicoScenes tool is used for CFR collection
- 360° video capture and CFR collection is synchronized
- Six different tests are performed in each of two different environment: anechoic chamber and hall room



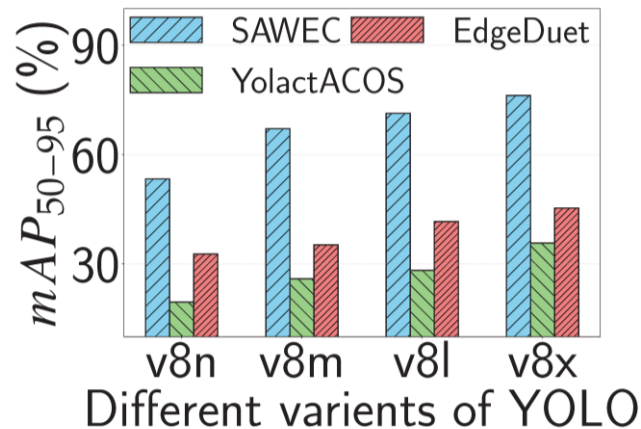


Instance segmentation  $mAP_{50-95}$  VS channel occupation at different  $\alpha$ :



- At  $\alpha = 2$  the ROI is sufficiently large, and the  $mAP_{50-95}$  reaches 67% with YOLOv8m

Instance segmentation  $mAP_{50-95}$  with different variants of YOLOv8

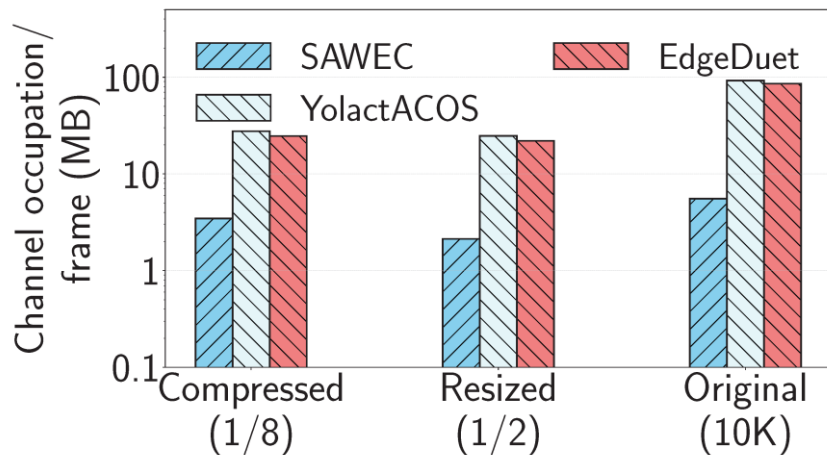


- YOLOv8x achieves an  $mAP_{50-95}$  of 76% while YolactACOS and EdgeDuet achieve only 36% and 45% respectively

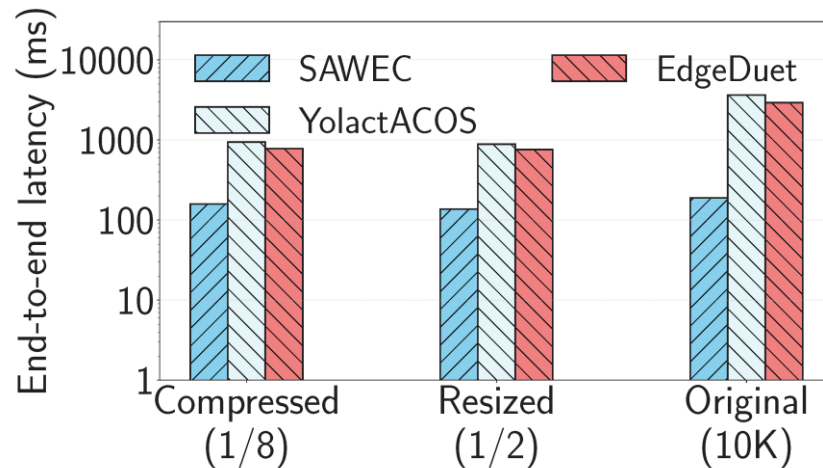
# Comparative analysis of SAWEC

Analysis with original (10K), and tolerable compressed (1/8) and resized images (1/2)

## Channel Occupation



## End-to-end Latency



- ❑ The size of a 10K frame is 138.88 MB
- ❑ SAWEC, YolactACOS, and EdgeDuet require 5.5 MB, 92.2 MB and 85.9 MB per frame respectively

- ❑ At 10K resolution, SAWEC improves the latency by 94% on an average

# Summary of Key Contributions

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- ✓ We have proposed SAWEC, a new paradigm for wireless edge computing
- ✓ E2E latency is optimized by offloading only the relevant data
- ✓ SAWEC reduces the channel occupation and E2E latency by 94% and 96%
- ✓ Improves the instance segmentation performance by 47%

# Possible Next Steps?



This  
work 💪

⚠️ Improve  
localization  
accuracy

Things  
to do 😄

⚠️ Scale the  
system to several  
moving objects

| **Thank You**



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